



Department of Statistics & Computer Science
University of Kelaniya
Academic Year -2024/2025 (Semester II)

COSC 32133/BECS 32263

Full-Stack Software Development

Lab Sheet 01

By the end of this lab, you will be able to:

Build a clean, responsive page layout using plain HTML and CSS, JavaScript.

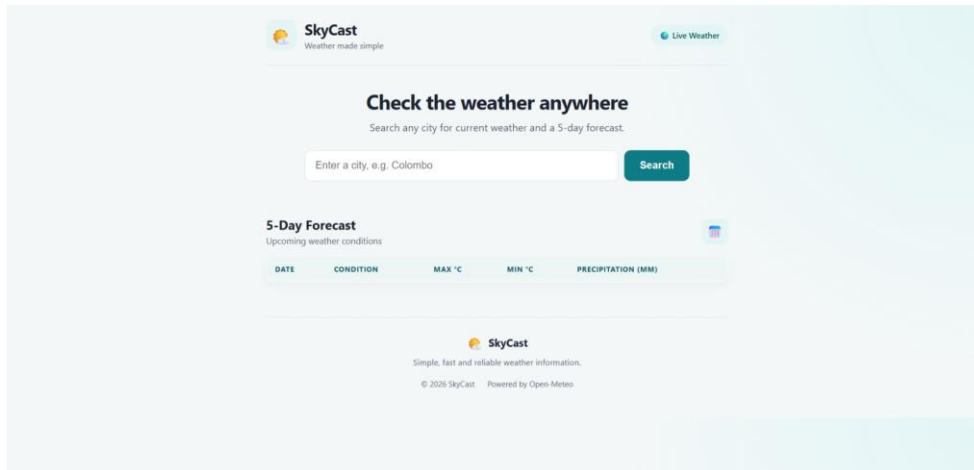
Consume a public REST API from the browser using the JavaScript `fetch ()` function.

Scenario:

SkyCast is a small weather-lookup website that allows users to enter the name of a city and click a Search button to view its current weather conditions. The page should display the current temperature, wind speed, and a short weather description such as “Partly cloudy.” In addition, a 5-day forecast table should display one row for each day, including the date, weather condition, maximum and minimum temperatures, and expected precipitation. All weather data must be retrieved from a live weather API through a JavaScript API call triggered when the user clicks the Search button. No back-end server is required for this lab.

Task 1 — Static Layout (HTML + CSS)

1. Create three files: `index.html`, `style.css`, `script.js`.
2. In `index.html`, build: a title/header for "SkyCast"; a search box with a text input and a **Search button**; a "current weather" card (empty placeholders for city name, temperature, wind, condition); an empty forecast `<table>` with header row Date, Condition, Max °C, Min °C, Precipitation (mm) and an empty `<body>`.
3. In `style.css`, style the page so it looks presentable: center the layout, style the card, buttons, and table, and make it responsive on a narrow screen.



Task 2 — Geocoding: City Name → Coordinates

1. In `script.js`, add a **click event listener** on the Search button.
2. Read the city name from the input and call the Geocoding API using `fetch()` inside an `async` function.
3. Parse the JSON response and extract `latitude`, `longitude`, and `name` from `results[0]`.
4. Handle the "not found" case: if `results` are missing or empty, show a friendly message instead of crashing.
5. Confirm your coordinates are correct with a `console.log ()` before moving on.

Geocoding: turn a city name into coordinates

```
GET https://geocoding-api.open-meteo.com/v1/search?name=Colombo&count=1

Response (shortened):
{
  "results": [
    { "name": "Colombo", "latitude": 6.9319, "longitude": 79.8478, "country": "Sri Lanka"
  }
  ]
}
```

Task 3 — Fetch and Display Current Weather

1. Using the coordinates from Task 2, call the Forecast API when the button is clicked.
2. Show a simple loading state (e.g. "Loading..." text) while the fetch is in progress and clear it once data arrives.
3. Update the current weather card with city name, `current_weather.temperature`, `current_weather.windspeed`, and a text description derived from `current_weather.weathercode`.

4. Wrap your fetch calls in `try/catch` so a network error also shows a friendly message.

Forecast: current weather + 5-day daily forecast

```
GET https://api.open-meteo.com/v1/forecast?latitude=6.9319&longitude=79.8478
    &current_weather=true
    &daily=weathercode,temperature_2m_max,temperature_2m_min,precipitation_sum
    &timezone=auto

Response (shortened):
{
  "current_weather": { "temperature": 29.4, "windspeed": 11.2, "weathercode": 3 },
  "daily": {
    "time": ["2026-09-16", "2026-09-17", "2026-09-18", "2026-09-19", "2026-09-20"],
    "weathercode": [3, 61, 61, 2, 1],
    "temperature_2m_max": [30.1, 28.4, 27.9, 29.0, 30.2],
    "temperature_2m_min": [24.5, 23.8, 23.5, 24.0, 24.7],
    "precipitation_sum": [0.0, 6.2, 4.8, 0.0, 0.0]
  }
}
```

Task 4 — Populate the Forecast Table

1. Loop over the parallel arrays in the `daily` object (`time`, `weathercode`, `temperature_2m_max`, `temperature_2m_min`, `precipitation_sum`) using their shared index.
2. For each day, create a new `<tr>` with five `<td>` cells and append it to the table's `<tbody>`.
3. Clear any previous rows before inserting new ones, so searching a second city doesn't duplicate rows.
4. Stretch goal: highlight any row where `precipitation_sum > 0` with a light blue background to flag rainy days and add a °C / °F toggle button.

Weather Codes

Code	Meaning	Code	Meaning
0	Clear sky	61 – 67	Rain
1 – 3	Mainly clear / partly cloudy	71 – 77	Snow
45, 48	Fog	80 – 82	Rain showers
51 – 57	Drizzle	95 – 99	Thunderstorm